Practical- 02

Title of the Assignment: A* algorithm for 8 puzzle problem

Problem statement: Implement A star Algorithm for any game search problem.

Objective:

- ➢ To understand the concept of Informed search techniques.
- ➢ To implement A* algorithm for 8 puzzle game problem.

Theory:

A* algorithm:

A* is a computer algorithm that is widely used in path finding and graph traversal, the process of plotting an efficiently traversable path between multiple points, called nodes. Noted for its performance and accuracy, it enjoys widespread use.

The key feature of the A* algorithm is that it keeps a track of each visited node which helps in ignoring the nodes that are already visited, saving a huge amount of time. It also has a list that holds all the nodes that are left to be explored and it chooses the most optimal node from this list, thus saving time not exploring unnecessary or less optimal nodes.

Here we use two lists namely 'open list 'and 'closed list'.The open list contains all the nodes that are being generated and are not existing in the closed list and each node explored after it's neighboring nodes are discovered is put in the closed list and the neighbors are put in the open list this is how the nodes expand. Each node has a pointer to its parent so that at any given point it can retrace the path to the parent. Initially, the open list holds the start (Initial) node. The next node chosen from the open list is based on its **f score**; the node with the least f score is picked up and explored.

f-score = h-score + g-score

A* uses a combination of heuristic value (h-score: how far the goal node is) as well as the g-score (i.e. the number of nodes traversed from the start node to current node).

- ➢ A* Algorithm is one of the best path finding algorithms.
- ➢ But it does not produce the shortest path always.
- ➢ This is because it heavily depends on heuristics.

Algorithm:

Step 1: Start.
Step 2: Define the initial state and goal state.
Step 3: Calculate the heuristic value h(n) using Manhattan Distance.
Step 4: Set:
[
g(n)=0
]
for the initial state.
Step 5: Calculate:
[
f(n)=g(n)+h(n)
]
Step 6: Insert the initial state into the priority queue.
Step 7: Select the state with the lowest f(n).
Step 8: If the selected state is the goal state, display the solution path and stop.
Step 9: Otherwise, generate all valid states by moving the blank tile.
Step 10: Calculate g(n), h(n) and f(n) for each new state.
Step 11: Add the new states to the priority queue.
Step 12: Repeat Steps 7–11 until the goal state is reached.
Step 13: Display the sequence of moves.
Step 14: Stop.

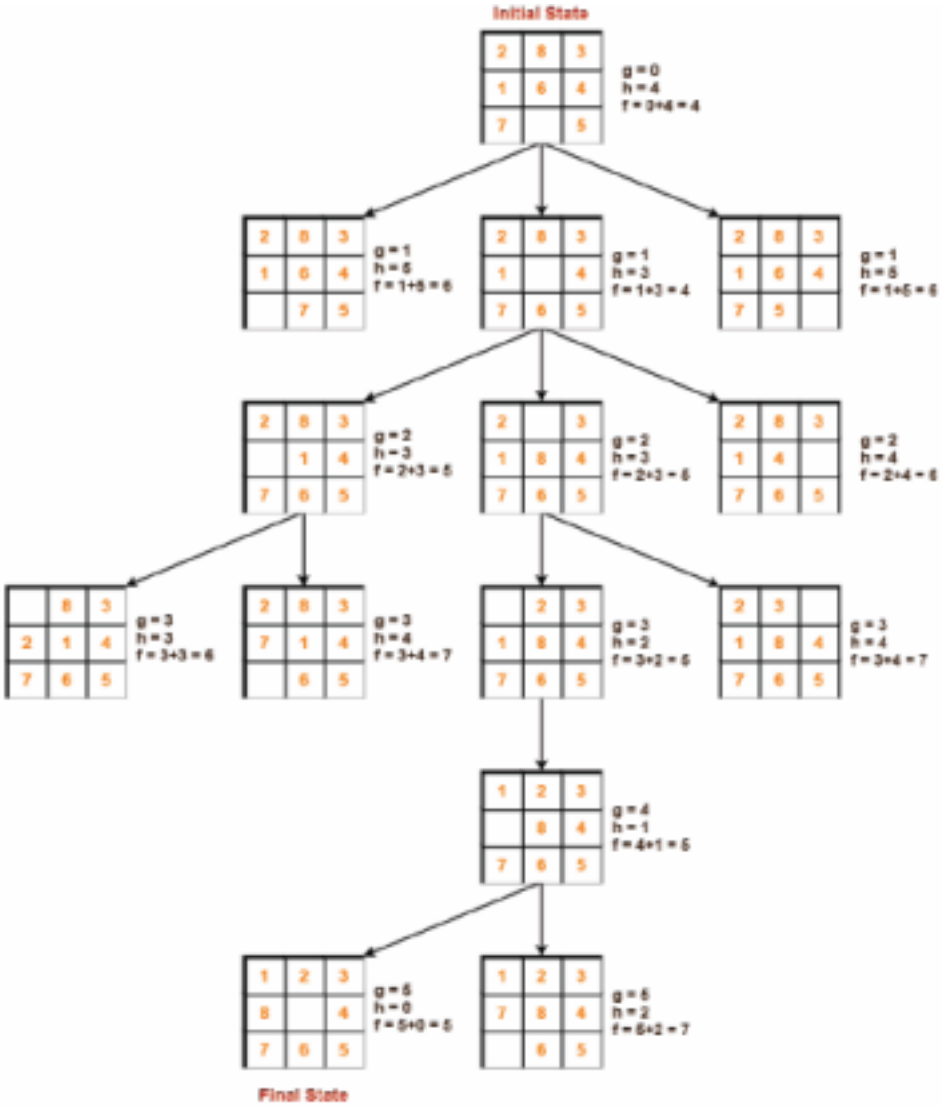
8-Puzzle problem:

In our 8-Puzzle problem, we can define the **h-score** as the number of misplaced tiles by comparing the current state and the goal state or summation of the Manhattan distance between misplaced nodes.
g-score will remain as the number of nodes traversed from a start node to get to the current node.
We can calculate the **h-score** by comparing the initial (current) state and goal state and counting the number of misplaced tiles.

Given an initial state of a 8-puzzle problem and final state to be reached-



Find the most cost-effective path to reach the final state from initial state using A* Algorithm. Consider g(n) = Depth of node and h(n) = Number of misplaced tiles.



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Laboratory Practice II

Conclusion:

Oral questions:

1. What are informed and uninformed search techniques?
2. List differences between informed and uninformed search techniques?
3. What is the evaluation function of A* algorithm?
4. What are the applications of A* algorithm?
5. What is problem solving agent?
6. Is A* algorithm informed or uninformed search technique?
7. What is Best first search algorithm?
8. Where are searching techniques applied in AI?
9. What are Blind search algorithms?
10. What is meant by Heuristic?